

Lecture 5 (9/23)

Expectation, Variance, and the Normal Distribution

Unit 1 Review

- Probability models: sample spaces, outcomes, events, and probability distributions
- Discrete random variables; joint, marginal, and conditional probabilities; Bayes' formula
- Independence and iid samples; simulation as an approximation to probability; translating between code and probability notation
- Continuous random variables: Uniform distribution, densities, CDFs

Goals for Unit 2:

- Expectation and variance, coefficient of variation
- Normal distribution; standardization and linear transformations
- The linear regression model assumptions
- Understand covariance and correlation

Sample averages and expectation

- The sample mean of iid draws $Y_1, \dots, Y_N$ of a variable with sample space S is

$$\overline{Y} = \text{np.mean}(y) = \frac{1}{N} \sum_{i=1}^N Y_i = \sum_{y \in S} y \frac{\#\{Y_i = y\}}{N}$$

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- For a continuous random variable with density $f_Y(y)$

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy$$

Expectation of a function

For any function h^1 of Y , we can similarly write

$$E[h(Y)] = \sum_{y \in S} h(y) P_Y(y) \quad \text{OR} \quad E[h(Y)] = \int h(y) f_Y(y) dy$$

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Taking $g(y) = 1_A(y)$ gives

$$E[1_A(Y)] = P(Y \in A).$$

A probability is just the expectation of an indicator.

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Example

Let Y take values in $\{1, 2, 3\}$ with $P(Y=1) = \frac{1}{2}$, $P(Y=2) = \frac{1}{3}$, $P(Y=3) = \frac{1}{6}$.
Compute $E[Y]$ and $E[Y^2]$.

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Example

Let $Y \sim \text{Uniform}(0, L)$. What is $E[Y]$? What is $E[Y^2]$?

Properties of expectation

- **Multiplication by a constant:** $E[aX] = aE[X]$
- **Factorization when independent:** If X and Y are independent, $E[XY] = E[X]E[Y]$.
- **Tower property:** $E[E[X | Y]] = E[X]$ where $E[X | Y]$ is the expectation of $X|Y$ (and hence depends on Y). This is just marginalization for expectations.

In each of the examples below, you should also understand how to write code which simulates the model and test the formula with these simulations.

Example

Let $X \sim \text{Bernoulli}(1/2)$ and suppose $Y \in \{1, 2, 3\}$ with conditional distribution

	$Y=1$	$Y=2$	$Y=3$
$P(Y X=0)$	$1/2$	$1/4$	$1/4$
$P(Y X=1)$	$1/3$	$1/3$	$1/3$

Compute $E[Y]$ and $E[XY]$.

Example

Let

$$\begin{aligned}X &\sim \text{Uniform}(0, 2) \\ Y|X &\sim \text{Uniform}(0, X)\end{aligned}$$

Compute $E[Y]$ and $E[XY]$.

Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2] = E[Y^2] - E[Y]^2.$$

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Example

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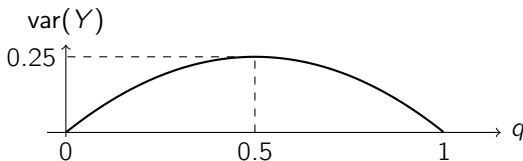
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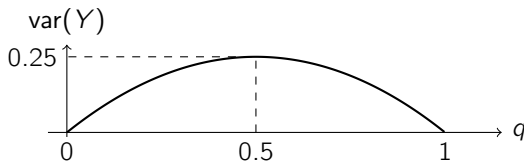
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Example

Let $Y \sim \text{Uniform}(0, 1)$. Compute $\text{var}(Y)$.

Standard deviation

- The standard deviation is the square root of the variance,

$$\sigma = \sqrt{\text{var}(Y)} = \sqrt{E[(Y - E[Y])^2]}.$$

Unlike the variance, σ has the same units as Y , so it tells us the *typical size* of the residual $Y - E[Y]$.

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- The coefficient of variation, or CV , compares the standard deviation to the mean:

$$CV = \frac{\sigma}{\mu}$$

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Quantity	Typical CV
Adult height	≈ 0.05
Adult weight	≈ 0.2
Household income	≈ 1
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Note: The CV makes sense for positive random variables. For example, it is meaningless for something like temperature.

The Normal distribution

$Y \sim \text{Normal}(\mu, \sigma^2)$ has density

$$g(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/2\sigma^2}, \quad E[Y] = \mu, \quad \text{var}(Y) = \sigma^2.$$

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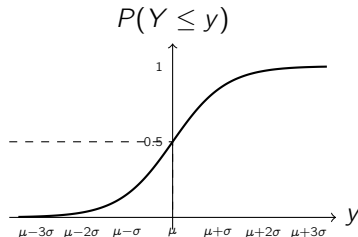
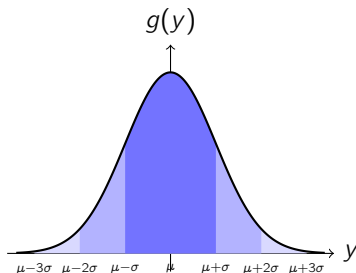
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- If X_1, X_2 are independent Normal and a, b, d are constants, then

$$aX_1 + bX_2 + d \sim \text{Normal}(a\mu_1 + b\mu_2 + d, a^2\sigma_1^2 + b^2\sigma_2^2)$$

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Example

Suppose $Y \sim \text{Normal}(20, 25)$.

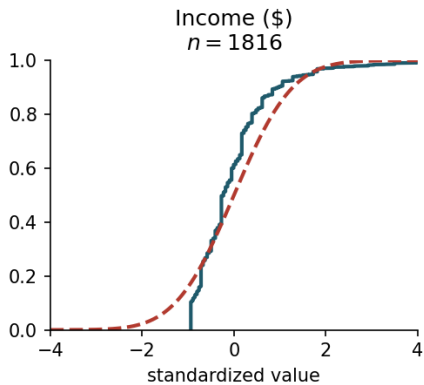
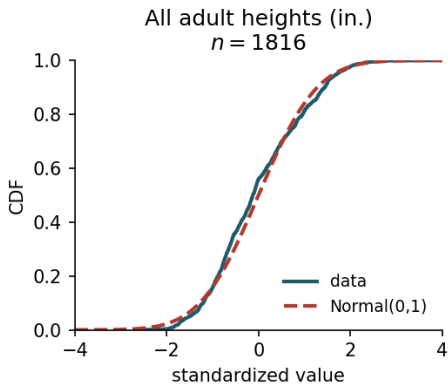
- Estimate $P(Y > 25)$ using the bell curve shape.
- Estimate $P(15 < Y < 30)$.

Plotting the CDF from samples

```
import numpy as np

y = np.random.normal(0, 1, size=10000)

# CDF estimate at each threshold t
dt = 0.01
ts = [-4 + i * dt for i in range(800)]
cdf_hat = [len(y[y < t]) / len(y) for t in ts]
```



Solutions

- 1) $Y \in \{1, 2, 3\}$, $P(Y=1)=\frac{1}{2}$, $P(Y=2)=\frac{1}{3}$, $P(Y=3)=\frac{1}{6}$: $E[Y] = \frac{5}{3}$, $E[Y^2] = \frac{10}{3}$.
- 2) $Y \sim \text{Uniform}(0, L)$: $E[Y] = \frac{L}{2}$, $E[Y^2] = \frac{L^2}{3}$.
- 3) $X \sim \text{Bernoulli}(\frac{1}{2})$, $Y | X$ from the table: $E[Y] = \frac{15}{8}$, $E[XY] = 1$.
- 4) $Y \in \{-1, 1\}$, $X | Y$ as above: $E[X | Y=1] = 0 = E[X | Y=-1]$, but not independent (e.g. $P(X=1 | Y=1) = \frac{1}{2} \neq 0 = P(X=1 | Y=-1)$).
- 5) $X \sim \text{Uniform}(0, 2)$, $Y | X \sim \text{Uniform}(0, X)$: $E[Y] = \frac{1}{2}$, $E[XY] = \frac{2}{3}$.
- 6) $Y \sim \text{Bernoulli}(q)$: $\text{var}(Y) = q(1 - q)$.
- 7) $Y \sim \text{Uniform}(0, 1)$: $\text{var}(Y) = \frac{1}{12}$.
- 8) $Y \sim \text{Normal}(20, 25)$: $P(Y > 25) \approx 0.16$, $P(15 < Y < 30) \approx 0.815$.